

Novel concepts for agent-based population modelling and simulation: GEPOC ABM 2025

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ABSTRACT

In recent years, dynamic agent-based population models, which model every inhabitant of a country as a statistically representative agent, have been gaining in popularity for decision support. This is mainly due to their high degree of flexibility with respect to their area of application. GEPOC ABM is one of these models. Developed in 2015, it is now a well-established decision support tool and has been successfully applied for a wide range of population-level research questions ranging from healthcare to logistics. At least in part, this success is attributable to the continuous improvement and development of new methods. While some of these are very application- or implementation-specific, others can be readily transferred to other population models. The present work focuses on the presentation of three selected transferable innovations that have had a major impact on the model in recent years: We illustrate an innovative time-update concept for the individual agents, a co-simulation-inspired simulation strategy, and a strategy for more accurate model parametrisation. We describe these methods in a reproducible manner, explain their advantages and provide ideas on how they can be transferred to other population models.

1. Introduction

With the continuous increase in computing power, agent-based (AB) population models are becoming increasingly popular as decision support tools. In these models, every agent statistically represents a real person. These agents can be assigned arbitrary attributes and behaviour patterns, and they can be tracked dynamically over any period of time.

These features give AB population models an almost limitless range of potential applications. Arguably, the most prominent field of application is epidemiology. Due to the microscopic nature of AB epidemic models, they allow consideration of details of the real system which cannot be depicted by classical compartmental models [1]. As a result, they have been heavily used for advising policymakers during the COVID-19 pandemic [2,3]. Other applications include the economy [4], opinion dynamics [5], and ecology [6]. Finally, one must not forget potential non-dynamic applications. A snapshot of the agent population in a population model can be regarded as a synthetic census, which can be used for arbitrary statistical analysis — a strategy which comes with various benefits [7] and drawbacks [8] compared to actually collected micro-/aggregated census data.

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Table 1

Overview of selected population models and related frameworks.

System	Country	Application	Time model	Interaction
STATSIM [21]	AT	Pop. projection	Continuous	No (case-based)
Demosim [22]	CA	Pop. projection	Continuous	No (case-based)
DYNASIM3 [27]	US	Retirement/ageing	Discrete (annual)	Limited
Destinie 2 [23]	FR	Pension/fiscal	Discrete (annual)	Limited
MOSART [28]	NO	Labour/pensions	Discrete (annual)	Limited
GEMS [24]	DE	Epidemic modelling	Discrete (tick)	Supported
LIAM2 [25]	Generic	Generic microsim.	Discrete (periodic;often annual)	Limited
GEPOC ABM	AT	Multi-purpose pop. ABM	Hybrid	Supported

In short, an agent-based population model can be a highly flexible and valuable tool for decision support. However, the significance of the model results is strongly linked to how accurately the real population and its dynamics are represented in the model. This requires highly accurate modelling and high-resolution parametrisation. Moreover, the sheer size of the agent population requires a highly efficient implementation to get model results in a reasonable computation time.

In 2015, a team at TU Wien and dwh GmbH started the development of their own Generic Population Concept (GEPOC), a collection of models and methods for multi-purpose population modelling and simulation [9], which serves as a sustainable model base for answering any kind of population-level research question. The most successful and most frequently applied of the models is GEPOC ABM, a generic agent-based population model [10]. The model is equipped with a default parametrisation for Austria, but could be applied to any country, given that the required data is available.

In the past, GEPOC ABM has been applied in a wide range of contexts. These include analyses of vaccination rates, re-hospitalisation of psychiatric patients, and the number, severity, and diagnosis of stroke incidence (we refer to [10] for details). Between 2020 and 2023, GEPOC ABM was heavily used for decision support during the COVID-19 crisis. Based on previous work on influenza, an epidemiological application of GEPOC ABM resulted in a highly flexible epidemic model, which became a core element of a COVID-19 model-family [11]. This model family was used to compute various what-if scenarios and forecasts related to containment policies [12], vaccination strategies [13], detection rate [14], and reporting analysis [15]. Due to the great flexibility of the model, it became one of the key elements in Austria's pandemic management. It was one of the three models harmonised within the Austrian Forecasting Consortium [16] and was one of the approaches in the ECDC European Scenario Hub. At the moment, the model is applied for analysis and economic comparison of 15 different vaccination programmes, decision support for climate-change-induced disasters such as flooding events [17], analysis of the human papillomavirus (HPV) disease dynamics, and pandemic preparedness [18].

1.1. Related work

It is quite difficult to compare GEPOC ABM with models in the international literature. Table 1 gives a selective overview of well-known and applied individual-based population models and related frameworks. Note that the table is not intended as an exhaustive taxonomy but mainly to underline the uniqueness of the approach. For broader surveys, we refer to [19,20].

This list can roughly be clustered into three classes. In class [a], we find time-continuous microsimulation models for the sole purpose of population forecasting, such as STATSIM [21] and Demosim [22]. Since these models neither support nor need agent interactions, they update in a case-based manner, i.e. each agent is updated individually, one after another. Class [b] consists of highly domain-specific models, built for a particular country and policy question, e.g. Destinie 2 [23] for pension projections, or GEMS [24] for epidemiological policy-making. Finally, we find class [c] with generic tools like LIAM2 [25]. All three classes differ from GEPOC ABM. First, GEPOC ABM is not updated in a case-based manner and is not intended for population forecasting, distinguishing it from class [a]. In contrast to the models from class [b], GEPOC ABM does not focus on a specific field of application. Finally, the generic tools from class [c] differ from GEPOC ABM by operating at the level of a simulator rather than a model. Furthermore, GEPOC ABM stands out because none of the models/tools from classes [b] and [c] supports any version of continuous-time update. While theoretical concepts for continuous-time agent-based population modelling have existed for many years (e.g. [26]), they are hardly ever used for real-world applications.

1.2. Goals

With ever-changing data and requirements, a tool like GEPOC ABM needs to be regularly revised, revalidated and refactored. This process spans everything from implementation and data-processing to the core of the tool itself: the conceptual model. We found that models like GEPOC ABM must meet several methodological challenges.

- **Flexibility.** A tool such as GEPOC ABM must be flexible with respect to potential applications. Therefore, the conceptual population model must be lightweight, intuitive, and easy to extend. In particular, event-driven updates (continuous-time agent-based modelling) must be supported for applications in which the modeller deals with non-trivial waiting times, e.g. state transitions.

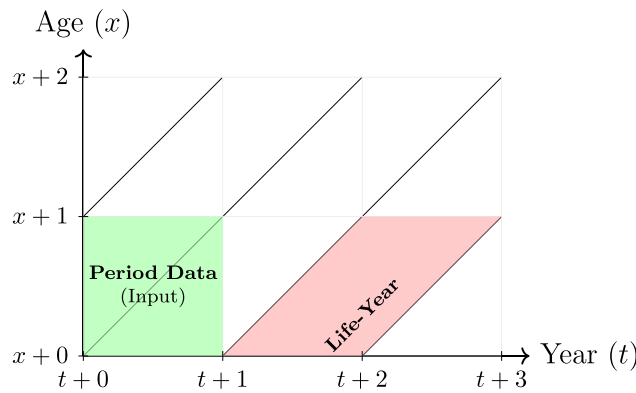


Fig. 1. Visualisation of the *chronometric dissonance* problem. Census data is reported for periods (green); individuals age along life-years (red). This means that whether or not a demographic event such as birth, migration or death will be counted in a specific census entry for an age-class depends on when in the course of the year it occurs.

- **Performance.** At the same time, the model must also be quick to execute, to avoid bottlenecks for decision support. This problem is often only discussed in terms of implementation; however, it affects model conceptualisation to at least the same degree. For example, using an event-based update strategy requires keeping some sort of event queue sorted all the time throughout the simulation. Independent of the algorithm, the necessary computational effort will be super-linear with respect to the number of events and consequently also the number of agents — typically millions in AB population models. Moreover, since any event might change any other future event, options for parallelisation are limited as well.
- **Accuracy.** The model must come with a maintainable parametrisation, which is quantitatively validated against reference data. At first glance, census data seems to be a straightforward source for parametrisation of agent behaviour; however, the devil lies in the details as soon as a certain level of accuracy and/or a high level of resolution is required. A central challenge in dynamic modelling is the *chronometric dissonance* between the administrative time of the state (calendar years) and the biological time of the individual (continuous ageing). As illustrated in Fig. 1, official demographic inputs are typically available as *period–age* statistics (represented by the **green square**): events observed in calendar year t and attributed to people of age x . However, agents become older along diagonal lifelines as illustrated by the **red parallelogram**: a simulation step tracking a biological life-year ($x \rightarrow x+1$) cuts across calendar boundaries. This creates a geometric mismatch: the “Life-Year” shape used by the model does not align with the “Period-Year” shape provided by the data. Any model must therefore choose a strategy to map these rectangular inputs to diagonal individual exposure.¹

Since 2018, when GEPOC ABM was most recently published [10], several advances have been made, which have rendered the tool more versatile and precise. Some of them are very specific to our implementation (e.g. the switch from Python3 to ABT/Java in 2019 [29]) or the available data formats for Austria, the country for which GEPOC ABM is most often used.

In this work, however, we want to present and discuss the most relevant and fundamental advancements: an improved update concept for the top-level simulation, a new conceptual model for the individual agents, and an improved parametrisation strategy. With these improvements, we address all mentioned modelling challenges at the same time and have made our GEPOC ABM not only more flexible, computationally efficient and accurate, but also more intuitive and stable. As proof, we will demonstrate its quantitative validity and present performance and accuracy benchmarks. We argue that the ideas can be well generalised to any large-scale AB population model and give modellers a new perspective for model conceptualisation.

In the following, Section 2 presents the innovations related to conceptual modelling, roughly explains how the model is implemented (Section 2.2) and gives examples of how the model is applied and extended for answering population-level research questions (Section 2.3). Furthermore, Section 3 focuses on data and explains the innovations related to computation of model parameter values. In Section 4 we describe the validation process of the model and present the results of the validation experiment (Section 4.2) and several benchmark experiments (Section 4.3). Finally, model, innovations, limitations and results are discussed in Section 5. For readers who are interested in understanding all details, we refer to [30] for full model documentation and to [31] for parameter documentation. These two reports were made publicly available via arXiv and include all formal details related to modelling, parametrisation and validation.

¹ Note that this problem did not exist, for example, under the traditional Korean age-counting system, in which all people simultaneously aged by one year on New Year's Day; South Korea officially replaced this system with the international age-counting system in 2023.

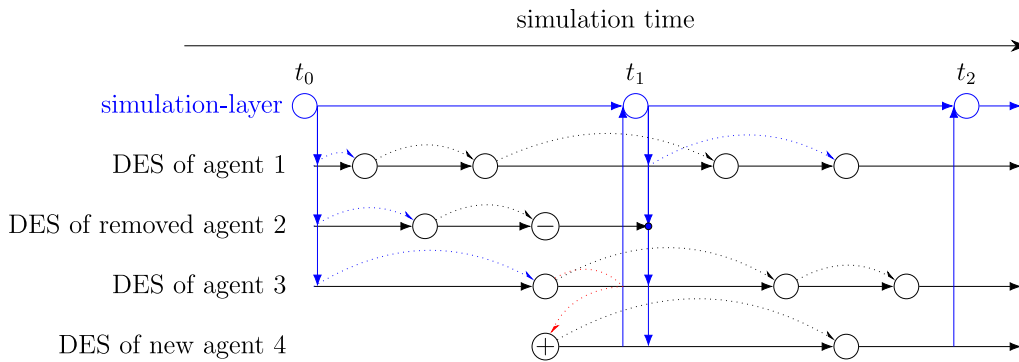


Fig. 2. Overall update scheme of GEPOC ABM. The time advancement is sketched with solid horizontal arrows, event-scheduling with dotted harpoons. The simulation-layer acts as runtime-infrastructure for the discrete-event simulations (DESs) of the individual agents. In every macro step, the simulation-layer (blue) first observes the individual states of the DESs (upwards arrows), and potentially interferes with them (downwards arrows), which includes adding or removing events (blue dotted arrows) or entire agents. The red dotted arrow displays an interaction between agents via the simulation-layer — in this case, a *Birth* event.

2. Modelling methods

2.1. Conceptual model

The conceptual model is best described using two layers: the simulation-layer, where agent interaction is modelled, and the agent-layer, where the agents' dynamics are modelled.

Note that for the presentation of the conceptual model, we will focus on clear communication rather than formal correctness and completeness. For the latter, we refer to [30].

Simulation-layer. To make an AB population model with millions of agents computable with an event-based update, we specified a very specific update strategy, which can best be described as a form of co-simulation [32]. In this approach, each agent follows its own individual discrete-event simulation (DES), which is harmonised by a synchronisation reference, henceforth called the *simulation-layer*. As in traditional parallel co-simulation, the simulation-layer uses macro steps on which it synchronises the individual DESs (see Fig. 2). At every step, the simulation-layer first observes the individual DESs and tracks whether an agent (a) experienced an event which renders it *removed* in the course of the last macro step, (b) scheduled an event which should lead to the creation of a new agent, or (c) scheduled an event which addresses another existing agent. In case of (a), the simulation-layer would remove the corresponding agent from the harmonisation loop; in case of (b), it would initialise a new agent and the corresponding DES and would trigger its initialisation event. In case of (c), it would insert the event into the event queue of the other agent.² While this strategy leaves interaction between agents restricted to the macro steps, the individual agents themselves can be updated continuously in time. However, with a proper choice of the macro step size, this limitation can be overcome, depending on the application. Reasonable values range from “days” for epidemiological applications, to “years” for applications in environmental sciences. As compensation for the limited interaction possibilities, one obtains a model that combines discrete and event-based updates, evaluates quickly, and, as a bonus, opens a direct path towards parallelisation on the agent level. We refer to the Discussion section for details.

Finally, we want to emphasise that the simulation-layer in GEPOC ABM is also implemented as its own DES, where macro steps are simply implemented as events scheduled iteratively. This is necessary for using arbitrary non-equidistant time steps such as “months” or “years”, and enables a more modular structure when adding additional top-level events, e.g. for observation or runtime plotting.

Agent-layer. The agents themselves are modelled as traditional discrete-event simulations with extended perception of their environment, i.e. the other agents. For modelling demographics, the internationally acknowledged definition of the term *probability of death*, as it is used by various national census institutes, was the key inspiration.

Definition 2.1 (Probability of Death). “The probability of death at some age x refers to the probability of a person living until the age of x to die during that year of age”. (Online glossary of Statistics Finland [33])

The most interesting feature of this definition is how it is centred around an individual's year of life, and not around the calendar year. Since the definition is an international standard, there are various methods for its interplay with the population census, which

² Note that removal and addition of new simulation entities is usually not an element of traditional co-simulation.

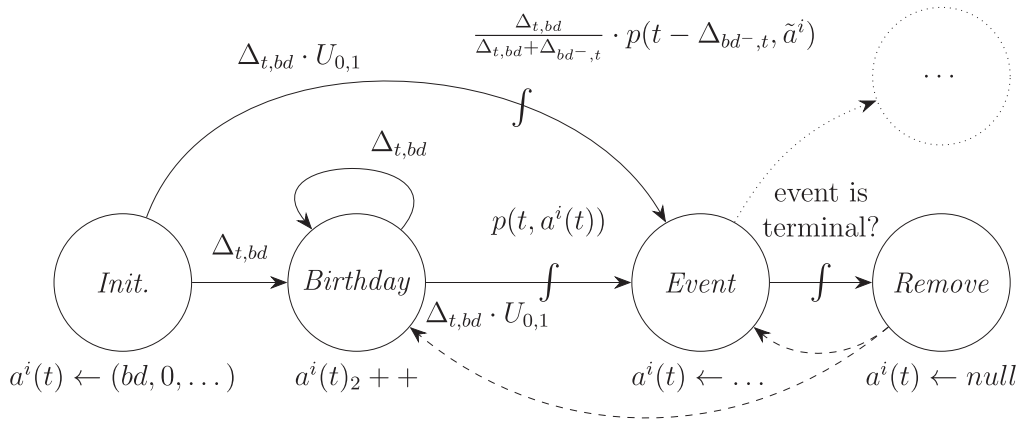


Fig. 3. Event graph representation of the birthday-centred update scheme for agents. A *Birthday* event regularly schedules itself every year. On its occurrence, demographic events (e.g. *Event*) taking place in the course of the agent's next life-year will be planned. They will be scheduled with a certain probability at a random point within the next life-year.

can be used for parametrisation (see below). Consequently, we defined an event-based update scheme built on the same principle, which we explain in the following paragraphs.

Let a^i stand for model-agent i and let $a^i(t)$ stand for the agent's state vector at time t . Here, $(a^i(t))_1 := bd$ refers to the static birthdate of the agent, $(a^i(t))_2 := age$ refers to its age in years, and $(a^i(t))_j, j = 3, \dots, n$ refer to other static or dynamic features of the agent such as sex, region/place of residence, annual income, or co-morbidities. In order to present the conceptual update scheme, we do not need to specify them any further.

With this definition, the update of the agent follows the Event Graph depicted in Fig. 3 (see [34] for a specification of the Event Graph formalism). The *Birthday* event is the core of the update scheme. It iteratively schedules itself every year on the recurrence of the agent's birthday and increments the agent's age (i.e. $a^i(t)_2$) by one. Here, in the graph, $\Delta_{t,bd}$ always refers to the time between the current time t and the corresponding next (re-)occurrence of the agent's birthday. It should be noted that this computation is not necessarily trivial, since "1-year" is not a proper unit, primarily due to leap years.

Upon occurrence of the *Birthday* event, time t - and state $a(t)$ -dependent probabilities decide if a demographic event such as death, birth, migration, etc. should occur within the upcoming life-year. In Fig. 3, this is illustrated by an example event *Event*, which occurs with probability $p(t, a^i(t))$. If the event is to be scheduled, it will be planned for a random point in time between the current and next *Birthday* event. This means that a time delay is computed by multiplication of $\Delta_{t,bd}$ with a random number $U_{0,1} \sim U(0, 1)$. We want to emphasise that the *Birthday* event also increases the agent's age by one. That means that the agent's age in years does not have to be (re-)computed from the difference between the current year and the agent's day of birth – as would be standard in other implementations – but comes "for free" as a dynamic agent state.

When a demographic event occurs, it can change the state $a(t)$, e.g. an *InternalMigration* event changes the place of residence. The event itself can also trigger further events, e.g. a *Birth* event initialises a new agent. This is indicated by the dotted scheduling edge in the picture. If an event is terminal, e.g. *Death*, the *Remove* event will be scheduled, which (a) renders the agent inactive, which is indicated in the Event Graph by setting the state to *null*, and (b) cancels all further planned events, indicated by the dashed cancelling edges.

In the model, a new agent is created when the initial agent population is initialised, when an agent "gives birth" to a new agent at runtime, or when an agent enters the system via immigration processes. In all three cases, an *Init* event sets up the initial state of the agent and schedules a *Birthday* event with delay $\Delta_{t,bd}$. Note that this timespan will be precisely one year if the agent was born at runtime, because the agent's creation date is identical to its birthdate. In all other cases, it will likely be shorter, because the agent's creation date t usually lies somewhere between two consecutive birthdays. However, in all cases, there is a chance that a demographic event will take place before the agent "celebrates" its first birthday at simulation runtime. The corresponding likelihood is, however, scaled down by $\Delta_{t,bd} / (\Delta_{t,bd} + \Delta_{bd^-,t})$. Here, $\Delta_{bd^-,t}$ refers to the time between the agent's last birthday and its creation date t , i.e. $\Delta_{t,bd} + \Delta_{bd^-,t}$ is precisely the length of the agent's current life-year. Note that we consider the day of birth as the 0th birthday in this context. There is, however, the problem that the corresponding probability function p must also be evaluated on the day of the agent's last birthday $t - \Delta_{bd^-,t}$. This requires finding an extrapolation $\tilde{a}^i$ for the unknown $a^i(t - \Delta_{bd^-,t})$, e.g. by $\tilde{a}^i \approx a^i(t)$.

In the most fundamental version, GEPOC ABM uses three demographic agent events: *Death* and *Emigration* represent terminal events that remove the agent from the model. *Birth* leads to the creation of a new agent via the simulation-layer. Note that immigration is performed externally by a different agent (Interface Agent), which we will not discuss any further in this work, but refer to [30]. In a more advanced version, an *InternalMigration* event is also possible, which causes an agent to change its place of residence within the country. Incorporating internal migration improves model accuracy but increases complexity and computational cost. Moreover, since there are versions of GEPOC ABM which do not include any spatial dimension, modelling of internal migration is not always necessary (see below, Section 2.2).

It should be noted that GEPOC ABM differentiates between male and female agents from a biological perspective, since it is primarily used to differentiate reproductive capacity; that is, *Birth* events are only scheduled for female agents. This choice is also reasonable from the parametrisation perspective since it is the most common concept in census statistics. However, if a specific application requires the use of additional social genders, the agents must be extended by the corresponding attributes.

2.2. Model implementation

Since 2019, the model has been implemented in Java using the Agent-Based Template (ABT, see [29]). The framework offers sufficient degrees of freedom to support any application/extension of GEPOC ABM, while at the same time providing efficient default methods for things like random number generation, time/date conversions, IO-routines, serialisation and deserialisation of agents, and event queueing.

Each GEPOC application is started by a run class that reads one (or more) configuration files in JSON format and, according to the configured options, starts different parallel threads for Monte Carlo simulation and automatically summarises their results in statistics. In each thread, an instance of the actual simulation is instantiated, which represents the simulation-layer described in the conceptual model. It takes care of loading the parameter files, instantiating and handling the agent population, and generating simulation output via its own recorder classes. As usual in Java, agents, events, and the simulation itself are implemented as separate classes. To make the framework easily extensible, it uses class inheritance and method overriding in many places. This means, for example, that anyone who aims to extend GEPOC ABM for an application no longer has to worry about the basic demographic functionalities, because they only need to extend the already implemented base classes (base agents, simulation, etc.), which take care of handling the demographic actions.

2.3. Model application

As described, GEPOC ABM was not developed for the purpose of population forecasting, but rather for the robust answering of population-specific research questions. Accordingly, the application of the model to a specific question always primarily consists of a **model extension**. This means that additional properties and behaviours are added to the agents. Sometimes even new agent types must be added; sometimes the environment is extended.

To give the reader a better idea of what such an extension might look like – at least conceptually – we describe two example extensions, which are both simplified versions of actual real-world applications of GEPOC ABM.

Vaccination coverage rates. Each member state undertakes to report annually to the WHO on vaccination coverage rates for selected diseases. In addition to the challenge of collecting data on the administered doses, there is also a methodological challenge due to population dynamics: Vaccinated people age, die or emigrate, and for immigrants, an a priori vaccination status must be estimated. Since 2017, TU Wien, commissioned by the Ministry of Health, has used GEPOC ABM to calculate Austrian vaccination coverage rates against measles and polio [35].

For this application, the model uses annual macro steps and is extended by a vaccine-provider-agent. Moreover, the agent properties are extended by a list of received vaccine shots — the agent's own vaccination record. For every macro step, the provider agent has access to a predefined number of vaccine doses, which it aims to administer to person-agents with specific properties in the course of the upcoming macro step. Properties always include sex and age (or birth-year), but may also include vaccination prerequisites for vaccination programmes with multiple stages. Person-agents themselves follow a certain vaccination schedule, as defined by the authorities. That means, together with their *Init* event, they are given a number of *RequestVaccination* events added to their event queues which are scheduled for those points in their lives, at which they should get a vaccination shot. When triggered, the person-agent sends a request for a vaccine shot to the vaccination provider. This request is handled at the next macro step via the simulation-layer. According to the stock of available doses for this macro step, a certain number of these requests will be fulfilled, and the vaccination will be written into the vaccination record of the requesting agent.

For the real-world application, the model is slightly more complex than described. For example, the model implements a logic for handling leftover doses and unfulfilled requests, and uses origin-dependent vaccination probabilities to sample the initial vaccination status of immigrants. The model is still applied yearly and produces reliable results which can ultimately be found publicly on the WHO website.

Epidemiology of SARS-CoV-2. As mentioned in the Introduction, GEPOC ABM was heavily used for decision support in the COVID-19 crisis in Austria, where it was extended into an epidemiological model for analysing COVID-19 interventions [12]. This model extension is essentially based on the following logic.

First, every person-agent is extended by additional contact-, infectiousness-, disease-, and immunity-specific properties. In addition, the environment is expanded to include so-called *contact-locations*. These are passive agents that serve as contact points for person-agents and represent e.g. households, school-classes or workplaces. As soon as a person-agent is initialised, it is assigned to one or more of them. The model uses daily macro steps, in which every agent creates a number of unidirectional *Contact* events with other agents, which are randomly drawn from the assigned contact-locations. The events are parametrised with the infectiousness state of the agent and are put into the DE simulators of the contact partners via the simulation-layer. As soon as a contact partner executes a *Contact* event, it reads the infectiousness state of the contact partner from the event parameter and starts a random decision process to evaluate whether the contact causes a transmission. Besides personal characteristics, contact-location and seasonality, in particular the immune status of the executing agent and the infectivity of the contact partner play a role here.

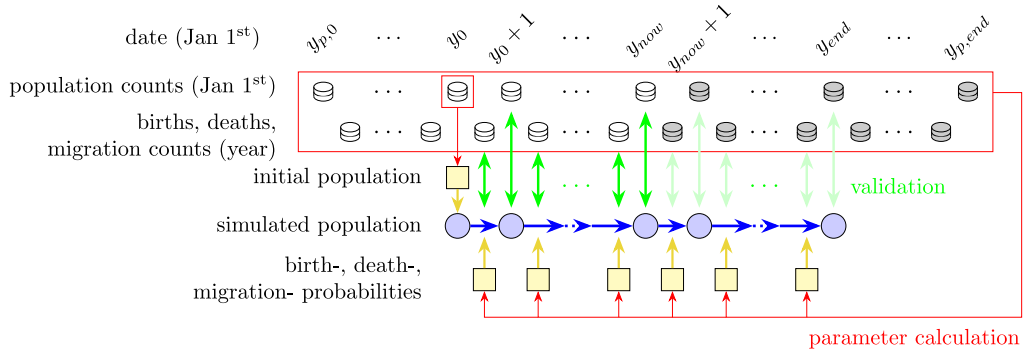


Fig. 4. General parametrisation and validation scheme of GEPOC ABM. Population, death, birth and migration counts for each year, in the form of published census counts (white) and forecasts (grey), serve both as the source of the model parameters (i.e. the initial population at the simulation start on January 1st in year y_0 , and the birth-, death-, and migration-probabilities for each simulation year) and as the reference for validation.

If the contact causes an infection, the agent enters a classic state chart, in which it passes through different stages of the disease. This means that as soon as the agent enters a disease-state, it draws a duration from a suitable positive distribution (exponential, gamma, etc.) and adds a corresponding *ExitState* event to its DE simulator. As soon as this event occurs, a decision process is triggered that determines which disease (or immune) stage(s) will be entered next.

The described concept outlines, in a nutshell, the purely epidemiological part of the SARS-CoV-2 model extension. Factors such as waning immunity or cross-immunity between different variants were also taken into account but have been omitted here for the sake of clarity. In order to use the model for the real-world analysis of interventions such as vaccinations, contact tracing or school closures, it was further extended by a comprehensive policy module. A complete description of all features is approximately 20 pages long and can be found in the supplement to [12].

3. Parametrisation methods

GEPOC ABM is only as valid as the parameters used, which are the size and distribution of the initial population, the number and distribution of immigrants entering the simulation at runtime, and the probabilities for demographic events mentioned earlier.

To enable the highest amount of flexibility, GEPOC ABM is based on semi-automatically updated parametrisation data, which is computed from openly available sources. For Austria, which is the main area of application of GEPOC ABM, the parametrisation data is computed from 14 different openly available datasets, including historical census and forecasting data for various demographic indicators, as well as geographical data (shapes and raster data) for regional sampling. The most important source of data is the official national statistics bureau Statistics Austria and its regularly updated Open Data platform <https://data.statistik.gv.at/web/>. As displayed in Fig. 4, the data is used as a source for a priori computation of the demographic parameters (probabilities) and for the initialisation of the initial agent population: Depending on whether the start date y_0 of the simulation is in the past, present, or future, the simulation is initialised either with the corresponding population counts from the census data or with the official population count forecast (small red box). From this point onward, the simulation runs independently of any population counts and uses only the parameters describing the demographic change, which are computed a priori from the published census data and forecasts (large red box). The parameters refer to the discussed individual probabilities for demographic events (deaths, births, etc.) but also the total number of yearly immigrant agents, for which the published immigrant counts and forecasts are used directly.

3.1. Data disaggregation and linkage methods

The most obvious challenge for parameter computation is the harmonisation of two datasets for two different timespans with different resolutions, for example, merging recent census data with very high spatial and age resolution with low-resolution forecasts. To solve the mismatch, we apply classic statistical data disaggregation onto the highest degree of resolution, using reasonable assumptions for the underlying distribution, usually taken from the temporally closest available high-resolution dataset.

For example, let $P(2025, [10, 19])$ refer to the population with age between 10 and 19 at 2025-01-01, and assume that data with high age resolution is available for 2024, i.e. $P(2024, [i, i])$ for $i \in \{10, \dots, 19\}$, then we may compute a distribution vector $\tilde{p}_i := \frac{P(2024, [i, i])}{\sum_{j=10}^{19} P(2024, [j, j])}$ and disaggregate

$$\forall i \in \{10, \dots, 19\} : \tilde{P}(2025, [i, i]) := P(2025, [10, 19]) \tilde{p}_i. \quad (1)$$

In some cases, the disaggregation must result in whole numbers, e.g. for the computation of the initial population size. For those cases, a variant of the Huntington–Hill method [36] is applied. This method was originally applied for fair distribution of parliament seats with respect to election results; however, the concept perfectly applies to disaggregation tasks with integer constraints.

Internal migration data turned out to be a special case of this problem. The official census data provides age-dependent counts for internal emigrants and immigrants, and inter-regional flows between the regions without age resolution. The three datasets can be interpreted as marginals of an unknown internal migration census, where the sum over one of the three dimensions, origin, destination and age, must result in one of the three datasets. In order to fulfil this disaggregation task, we apply three-dimensional Iterative Proportional Fitting (IPF, see [37]). Based on an initial guess, this algorithm iteratively scales its internal state-tensor by the three defined marginals. If the system has a valid solution, the algorithm converges after sufficiently many scaling steps. The resulting data can be used for an internal migration model, which preserves valid origin–destination flows and the correct age-distribution for internal immigrants. Note that the resulting origin–destination flows per age group are real numbers and not integers, and must therefore be understood as a distribution rather than a person count.

Finally, we want to emphasise that the resolution-matching tasks for the open-data parametrisation of GEPOC ABM can range in complexity from the mentioned disaggregation of age-classes into single age groups, to the computation of high-resolution yearly deaths or births from scalar demographic indicators like life expectancy or mean-fertility age.

3.2. Methods for computation of probabilities from census data

As mentioned earlier, the computation of probabilities from classical census data is always problematic. However, with the applied modelling strategy, in which the probabilities are based on Definition 2.1, we can make use of existing, validated methods for their computation. The most important of these originates in the computation of life tables and was developed by life-science pioneer William Farr (1807–1883).

To illustrate this, we take the computation of death probabilities as an example. Let $P(y, a)$ stand for the population per (turn of the) year y , and age a , and $D(y, a)$ for the overall number of deaths of a -year-old individuals in year y (we drop any other dimensions such as sex and region for the sake of readability).

It is clear that naive computation of the average rate of mortality $m = D(y, a)/P(y, a)$ already provides a reasonable approximation to the probability of death $p_d(y, a)$; however, it does not properly address the mismatch between census data year and individual life year. Note that roughly half of the deceased individuals who contributed to $D(y, a)$ were actually $a - 1$ years old at the beginning of year y . Farr's death rate formula (see [38], page 848) takes this problem into account:

$$p_d(y, a) \approx 1 - \frac{1 - \frac{1}{2} \frac{D(y, a)}{P_{avg}(a)}}{1 + \frac{1}{2} \frac{D(y, a)}{P_{avg}(a)}} = \frac{D(y, a)}{P_{avg}(a) + \frac{1}{2} D(y, a)}. \quad (2)$$

The formula assumes a (temporally) constant rate of mortality, a constant age-cohort size $P_{avg}(a)$, and neglects migration. The formula can be motivated by the idea that the average rate of mortality m computes the rate of death per lived year of life, but not per individual. Precisely $P_{avg}(a)$ years are lived by persons with age a in the course of the observed year; however, since every deceased individual will instantaneously be replaced by a new one (e.g. from the next age cohort) to maintain the constant size of the age cohort, those years are lived by $P_{avg}(a) + D(y, a)$ distinct persons. Clearly, the assumptions of the Farr formula are not valid for a real population: rates of mortality are not really constant, migration is omnipresent and age cohorts become smaller and larger during the year. However, these changes are usually a lot smaller than the total number of occurring deaths. Therefore, the formula is still clearly more accurate than the naive division shown before. We refer to the corresponding benchmark in Section 4.3 for an empirical test.

By default, all parameters are computed on different regional aggregation levels (country, federal states, districts, municipalities). First, this concept provides the highest possible level of flexibility with respect to the model application. For example, it is possible to completely neglect any regional resolution if not needed and save valuable computation time. Furthermore, the concept also enables the selection of a suitable regional resolution for every demographic process individually. For example, in Austria, emigration probabilities are regionally highly diverse, and therefore the data with the highest available regional resolution should be used. In contrast, it turned out to be not only sufficient but also beneficial to consider death probability parameter values on the federal state resolution, since the parameter computation is more stable due to the larger population and death counts.

Currently, the established parameter files cover the maximum time period between 1996 and 2101. For a full specification of the parametrisation process, we refer to [31].

4. Results

4.1. Validation strategy and scenario

GEPOC ABM was developed as a multi-purpose simulation tool to answer research questions at the population level. Accordingly, the model can be considered valid if the dynamics of the agent population represented in the model correspond to those in the real world. This means that the data used to calculate the model parameters also serves as reference values for the validation (solid green arrows in Fig. 4).

GEPOC ABM is also intended to deliver meaningful results for periods in the future. Here, the official population forecast for Austria, produced by Statistics Austria using the STATSIM microsimulation model [21], can be considered as the reference data. This model is explicitly designed for population forecasting and can therefore be regarded as the best possible source of information for the future dynamics of the Austrian population, and emulating the corresponding results can be regarded as the gold standard

for GEPOC ABM. Nevertheless, the comparison between GEPOC ABM results and forecast data is a cross-model validation rather than an actual validation and must therefore be interpreted differently.

As a validation scenario, we define 2000-01-01 as the start-date and 2050-01-01 as the end-date of the simulation and use GEPOC ABM to dynamically update the population of Austria. That is, the model uses the population census data for 2000-01-01, the death, birth and migration parameters for the years 2000 to 2024 and the corresponding forecasts between 2025 and 2049. We choose yearly macro steps and run the model under full consideration of internal migration (IM) — using the terminology of [30], we apply the Full-Regional IM Model, which uses internal migration parameters resolved by age, origin and destination. The probability parameters defined in the previous sections are regarded as year-, age-, sex-, and region-dependent, whereas different resolutions are used for the latter: birth- and death-probabilities are different for every federal state (9), migration-probabilities are different for every district (roughly 100). This validation scenario will be referred to as scenario **SC1** henceforth.

Over the 50 simulated years, the simulation develops its own synthetic census by counting the total number of inhabitants (on each January 1st) and the numbers of births, deaths, and emigrants each year. For comparison with the actual data, we observe the synthetic census for the federal states of Austria, and by 20-year age classes. Hence, the output of a simulation run is defined as the four functions

$$\begin{aligned} P^{sim} : \{2000, \dots, 2050\} \times \{AT-1, \dots, AT-9\} \times \{m, f\} \times \{[0, 19], \dots, 80+\} &\rightarrow \mathbb{N} : \\ (y, r, s, a) &\mapsto P^{sim}(y, r, s, a) \end{aligned} \quad (3)$$

for the total population on $y - 01 - 01$ and with $X \in \{B, D, E\}$,

$$\begin{aligned} X^{sim} : \{2000, \dots, 2049\} \times \{AT-1, \dots, AT-9\} \times \{m, f\} \times \{[0, 19], \dots, 80+\} &\rightarrow \mathbb{N} : \\ (y, r, s, a) &\mapsto X^{sim}(y, r, s, a) \end{aligned} \quad (4)$$

for total births, deaths and emigrants over the year y . Validation of immigration is not necessary, since immigrants are directly and deterministically sampled from the census data.

Since they are outputs of a stochastic simulation, all functions P, B, D, E must be regarded as random variables. To account for this uncertainty, 9 Monte Carlo runs have been found to be sufficient, so that the sample mean

$$\bar{X}^{sim}(y, r, s, a) := \frac{1}{9} \sum_{i=1}^9 X_i^{sim}(y, r, s, a), \quad (5)$$

for $X \in \{P, B, D, E\}$ and $X_i^{sim}(y, r, s, a)$ being the result of the i th simulation run, approximates the actual unknown mean $\mu(X^{sim})$ sufficiently well, at least for all (y, r, s, a) with $\bar{X}^{sim}(y, r, s, a) \geq 1000$. This was evaluated using a Gaussian error assumption (compare with [39]) and by investigating the quotient between sample standard-deviation and sample mean.

As displayed in Fig. 4, the reference X^{data} , with $X \in \{P, B, D, E\}$, is a composite of different open data and forecast sources, which mainly originate from the Austrian national census office Statistics Austria and its Open Data platform [40]. We refer to the corresponding section in [31] for details.

As a validation metric, we define the maximum and minimum relative deviation e_{max}, e_{min} for a given time interval $[y_1, y_2]$, region r , sex s and age a as

$$e_{max}(\bar{X}^{sim}, r, s, a) := \max_{y \in \{y_1, \dots, y_2\}} \left(\frac{\bar{X}^{sim}(y, r, s, a) - X^{data}(y, r, s, a)}{\max(1, X^{data}(y, r, s, a))} \right), \quad (6)$$

$$e_{min}(\bar{X}^{sim}, r, s, a) := \min_{y \in \{y_1, \dots, y_2\}} \left(\frac{\bar{X}^{sim}(y, r, s, a) - X^{data}(y, r, s, a)}{\max(1, X^{data}(y, r, s, a))} \right). \quad (7)$$

This validation metric, which, to be mathematically precise, is not even a metric but rather an error interval, was chosen in favour of classic errors like Root Mean Squared Error, because it does not smooth over fluctuations and maintains the sign of the difference. As a result, potential errors due to temporary fluctuations and systematic under- or overestimation remain visible and can be interpreted.

Since the interval between the 5 and 95 percent sample quantiles

$(Q^{0.05}(X^{sim}), Q^{0.95}(X^{sim}))$ can be regarded as an empirical 90% uncertainty interval (UI) for X^{sim} , so can

$$(e_{max}(Q^{0.05}(X^{sim}), r, s, a), e_{max}(Q^{0.95}(X^{sim}), r, s, a)) \quad (8)$$

for the maximum deviation, and analogously for the minimum deviation. While we will state this interval in the result tables, we will omit uncertainty bands in the result plots. The latter only display highly aggregated system components, for which the uncertainty is very low, and the corresponding uncertainty band would be almost invisible.

4.2. Validation results

In a nutshell, Fig. 5 shows the similarity of the simulated census from SC1 to census data and forecasts. The period between 2000 and 2025, in which the results are compared to the actual data, stands out with a particularly good fit. The result curves for the total population are nearly indistinguishable from the reference and within $\pm 0.2\%$, and the births and deaths are within $\pm 3\%$ of the reference. Emigrants show a slightly larger deviation, though only because of heavy annual fluctuations, which are

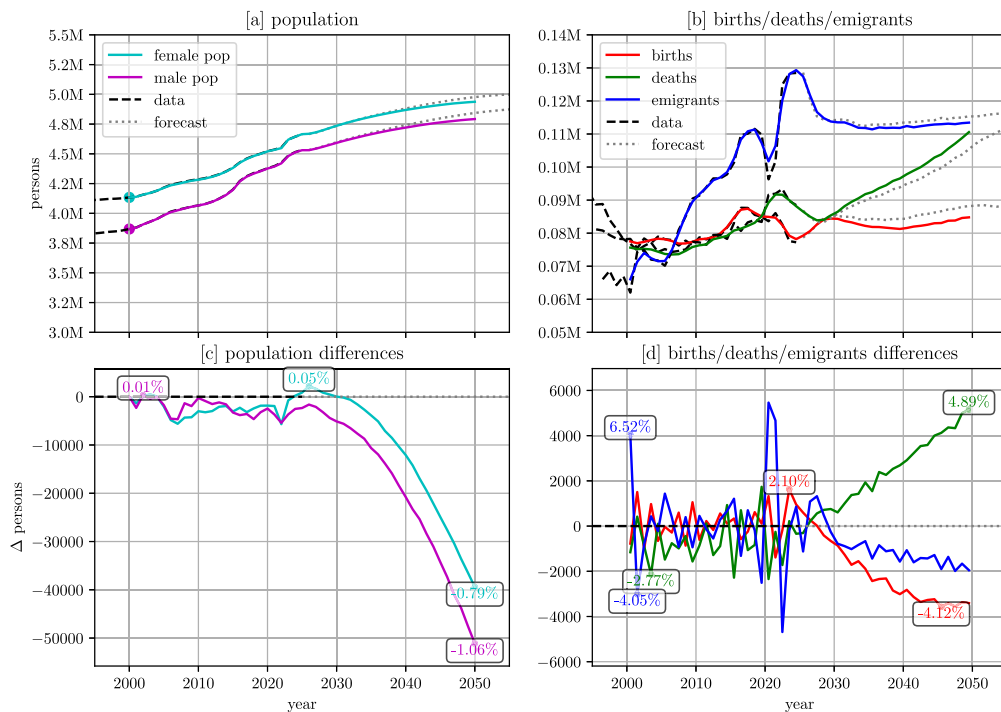


Fig. 5. Simulated census (full lines, simulation start in 2000) compared to the reference data (dashed: historical census counts up to year 2025, dotted: official forecasts from 2026 onwards) for different demographic indicators between 2000 and 2050. Panels [a] and [c] show the comparison of the population counts, differentiated by male and female, [b] and [d] show total births, deaths and emigrants. While panels [a] and [b] display total numbers, panels [c] and [d] show the differences together with the maximum and minimum deviations e_{max} , e_{min} .

slightly smoothed by the model. From 2025 to 2050, the results start to diverge from the reference forecast: births and emigrants tend to be underestimated; deaths are slightly overestimated. These differences are likely due to weaknesses in the approach for age-disaggregation of the low-resolution population, births, deaths and migration forecast data. However, they may also originate in structural differences between GEPOC ABM and the STATSIM model of Statistics Austria.

While panels [a] and [c] illustrate a near-perfect fit of the overall population until 2025, higher deviations are revealed when looking more closely at finer resolutions. Fig. 6 shows a detailed picture of the differences between the simulated and actual overall population for January 1st, 2025, with respect to sex, age-class and federal state. On this lower level of aggregation, the errors are larger, but with the exception of a few outliers for 20–39-year-olds in Salzburg (AT-5), they remain within $\pm 2\%$ after 25 simulated years. These outliers are due to a slight disruption of the migration in 2022, likely due to the COVID-19 crisis, which could not be fully depicted by the simulation and led to a temporary imbalance between the sexes in this region.

Finally, Tables A.3 and A.4 show the defined error metrics for the overall population on different levels of aggregation. As expected, relative differences tend to become larger with smaller-sized cohorts.

4.3. Benchmarks

In order to better analyse the benefits of the novelties, we defined a series of benchmarks. Some put the new GEPOC ABM version in contrast with the old one, while others explore the impact of different parameter computation strategies.

Computational performance benchmarks. First of all, we investigate the impact of the defined update scheme on the computational performance.

By design, the demographic simulation is independent of the choice of the macro step size. This is reasonable, since variation of it is motivated mainly by an application of GEPOC ABM to a specific problem requiring agent interaction.

The former GEPOC ABM version published in 2018 [10] was different in this regard. Demographic events were sampled for each macro step with scaled probabilities. This not only led to (marginally) different simulation outcomes when varying the macro step size (compare with Figure 3 in [9]), but also led to computational overhead. It should be noted that the level of accuracy achieved by the old version executed with the new data is of the same magnitude as the one for the new version — the smaller the macro step size, the more accurate. This highlights that the achieved accuracy is mainly a result of the good parametrisation routines.

Table 2 contrasts the computation times of the old and the new version of GEPOC ABM, using different macro step sizes ranging from daily to yearly. Both GEPOC ABM versions were run on a standard notebook (1.30 GHz, 12th Gen Intel® Core™ i5-1235U,

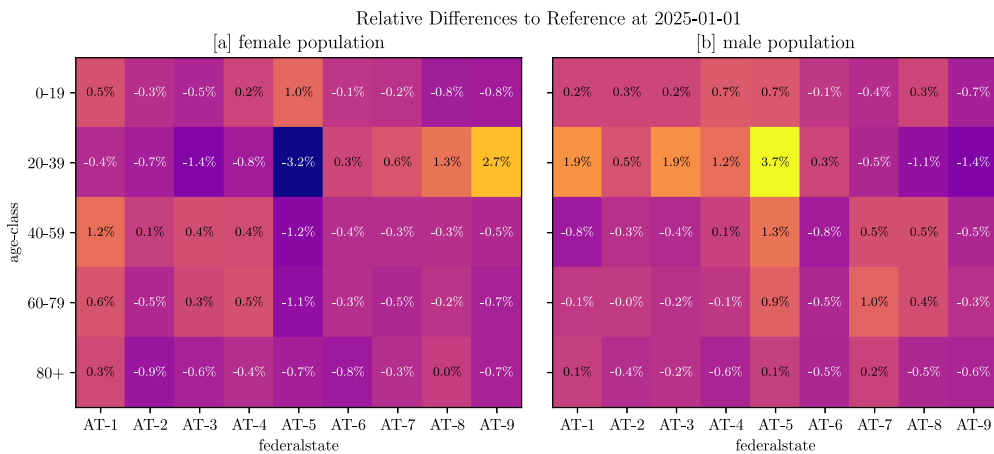


Fig. 6. Relative differences between simulated and reference male and female populations on January 1st, 2025, for different age-classes and the nine federal states AT-1 to AT-9.

Table 2

Computational performance comparison between the 2018 and 2025 versions of GEPOC ABM for different macro step sizes and scaling factors. The values in square brackets display the ratio to the first row.

scaling	macro step	# steps	GEPOC ABM 2018		GEPOC ABM 2025	
			duration (s)	$\frac{\text{duration}}{\text{step}}$ (s)	duration (s)	$\frac{\text{duration}}{\text{step}}$ (s)
1:100	years	50 [1.00]	23 [1.00]	0.460 [1.00]	12 [1.00]	0.245 [1.00]
1:100	months	600 [12.00]	271 [11.78]	0.452 [0.98]	24 [1.97]	0.040 [0.16]
1:100	weeks	2 609 [52.18]	937 [40.70]	0.359 [0.78]	51 [4.18]	0.020 [0.08]
1:100	days	18 263 [365.26]	6 204 [269.52]	0.340 [0.74]	260 [21.22]	0.014 [0.06]
1:1	years	50 [1.00]	3 932 [170.82]	78.6 [170.82]	979 [80.02]	19.6 [80.02]
1:1	months	600 [12.00]	–	–	1 855 [151.67]	3.1 [12.64]

upgraded to 48 GB RAM). Time for initialisation, e.g. agent-creation, and post-processing, e.g. Monte Carlo statistics computation, is not included. The major part of the comparison was computed with roughly 90 000 agents, i.e. scaled down by a factor of 1:100, due to time-constraints. The experiments in the bottom part of the table were run with full population to investigate how the computation time scales with the population size. Other configurations apart from step size and scaling are the ones from scenario SC1 (see Section 4.1).

In the old GEPOC ABM version, the runtime scales almost linearly with the number of macro steps. However, a day step is slightly shorter than a year step (approx. 74%), as fewer demographic events occur due to the lower probabilities. In the new GEPOC ABM version, the runtime also depends on the macro step, but this is solely due to reporting and additional iterations over the agent population. Therefore, the simulation with approximately 18 000 steps takes only about 20 times longer than the one with 50 steps.

When scaling things up to the full population of 9 million agents, one run with the new version and 50 yearly steps takes roughly 16 min. This is 80 times longer than the 1:100 scaled-down version, meaning that the computation time-demand increases roughly linearly with the population size, matching the behaviour expected from the model design. The computation time of the old version scales worse, since the 1:1 run is 171 times slower than the 1:100 run. We assume that this is due to Python-specific performance issues when handling very long lists and does not relate to the model structure which would also indicate linear scaling behaviour. Finally, it is worth noting that the factor of 1.97 observed for the 1:100 scaled new model version when changing from years to months is also correct for the full population ($1855/979 \approx 1.89$). Unfortunately, each run of the JAVA model requires almost 30 GB of memory meaning that the experiments were only just possible on mentioned laptop. As a result, we usually run the model on a suitable server with several hundred GB memory allowing parallelised Monte Carlo simulation on top. The memory demand of the old model version was slightly smaller (20 GB).

Accuracy benchmarks. Without a doubt, the parameter values underlying the model, which have been calculated from freely available sources using the approximation and aggregation methods described, are the decisive component in ensuring the accuracy of the model. In this context, one must of course ask whether the corresponding calculation methods are appropriate and precise.

It is clear that the probability of demographic events occurring is age- and region-dependent, and must therefore be integrated into a population model in this form — at least if the model claims to depict the population dynamics in age- and region-specific form. Since in particular (but not exclusively) forecast data is not available in age-specific form and often lacks the required regional resolution, additional assumptions for data linkage and disaggregation must be made in order to estimate the probabilities.

Internal migration is a good example to show the importance of this process: There are three publicly available datasets for internal migration in Austria, each only showing a part of the whole picture: (I) age-dependent internal emigrants per region, (II)

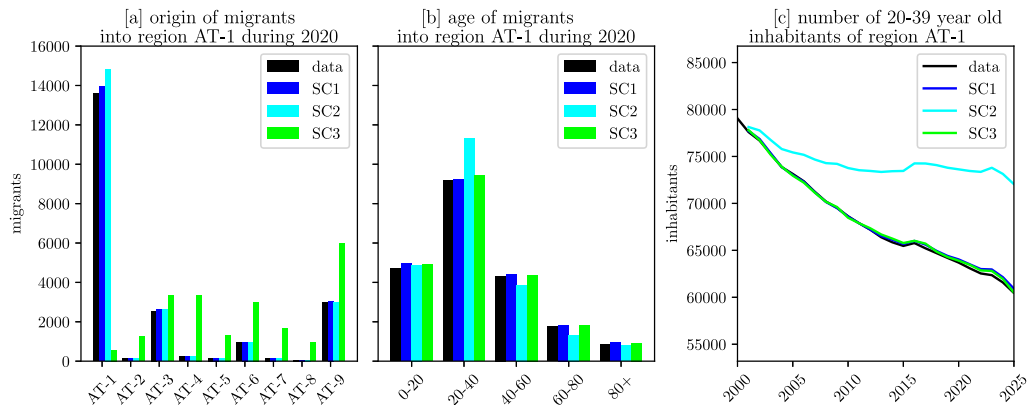


Fig. 7. Comparison between internal-migration scenarios SC1, SC2, and SC3 compared to data for selected census figures for Austrian federal state AT-1 (Burgenland). Panel [a] displays the total number of migrants into (or within) AT-1 per region of origin, panel [b] shows their age-distribution. Panel [c] displays the dynamics of the 20–39-year-old individuals in AT-1.

age-dependent internal immigrants per region, and (III) age-independent internal migrants per origin and destination. As mentioned, GEPOC ABM uses the IPF algorithm to link the three into one age-dependent dataset for migrants per origin and destination region.

We define two additional benchmark scenarios, in which internal migration is modelled using only two of the three datasets at once, without additional disaggregation. Scenario **SC2** samples internal migrants using dataset (I) and relocates them based on dataset (III), regardless of their age. Scenario **SC3** follows the same strategy for emigration, but relocates them using dataset (II), without considering their origin region.

Fig. 7 compares the three scenarios SC1–SC3 with the public data for federal state AT-1 (Burgenland). Panel [a] displays the simulated and reported internal migrants into AT-1 during the year 2020, with respect to the corresponding region of origin. SC3 does not reflect the real data at all, since destination regions are drawn only to fit the age-distribution of internal immigrants. Panel [b] displays the age-profile of the internal immigrants into AT-1. Here, SC2 does not fit the real situation well, since destination regions are drawn from an age-independent destination distribution. While age-distribution errors do not seem large at first, they accumulate over several years and lead to inaccurate demographic forecasts, which can be seen in panel [c]. Only SC1, which relies on statistically linked datasets with the IPF algorithm, is able to match the data from all perspectives, because the linked dataset fulfils all three boundary conditions.

Finally, the suitability of the proposed death rate formula must also be questioned. For this purpose, benchmark scenario **SC4** was established, in which probabilities are calculated using the naive formula for the average rate of mortality m (see Section 3.2) instead of the Farr formula (2). A comparison between SC1 and SC4 is displayed in **Fig. 8**, focusing on the period between 2000 and 2025. The naive parameter computation also leads to a reasonably well-fitting model; however, the errors for male and female populations in the observed period are two to three times larger. Also, the deviations for emigrants, deaths and births are much larger. **Fig. 9** is the analogue to **Fig. 5** for SC4.

5. Discussion

In this paper, we have presented several relevant updates to the agent-based population model GEPOC ABM. The concepts presented have contributed to making the model even more accurate, flexible and intuitive when applied for a specific population-level case study. Thematically, they range from conceptual modelling, implementation and simulation to model parametrisation, and we are confident that many of them can readily be transferred to other models:

Co-simulation-inspired time-dynamics. As seen from the performance benchmarks in Section 4.3, computation time scales linearly with the number of agents (15–20 min for 9 million agents vs. 12 s for 90 000 agents). This is a direct result of the presented co-simulation-inspired update scheme (**Fig. 2**) and is not self-evident for event-driven models: Increasing the number of agents linearly increases the number of events in the event queue, and a queue with N entries usually leads to a computational effort of magnitude $\mathcal{O}(N \log(N))$ for scheduling and polling — at least when using standard event queuing methods like heaps or trees [41]. Moreover, the scheme offers an intuitive approach for the parallelisation of population-dynamic, time-continuous, agent-based systems. However, this feature is not utilised in the current implementation of GEPOC ABM, mainly because all available computing resources can be fully utilised by the parallel execution of independent Monte Carlo simulation runs. The scheme should be understood as a deliberately narrower, domain-specific synchronisation strategy rather than a competitor to general-purpose parallel discrete-event simulation (PDES, [42,43]) or High Level Architecture (HLA, [44,45])-based approaches: it exploits the homogeneous structure of population-model agents instead of providing general distributed-DES protocols, and is largely complementary to those frameworks.

On the downside, interaction between agents is limited to the macro steps, which makes GEPOC ABM not a fully time-continuous agent-based model. Even though this has not yet posed a problem for us in previous applications – so far, all questions have been

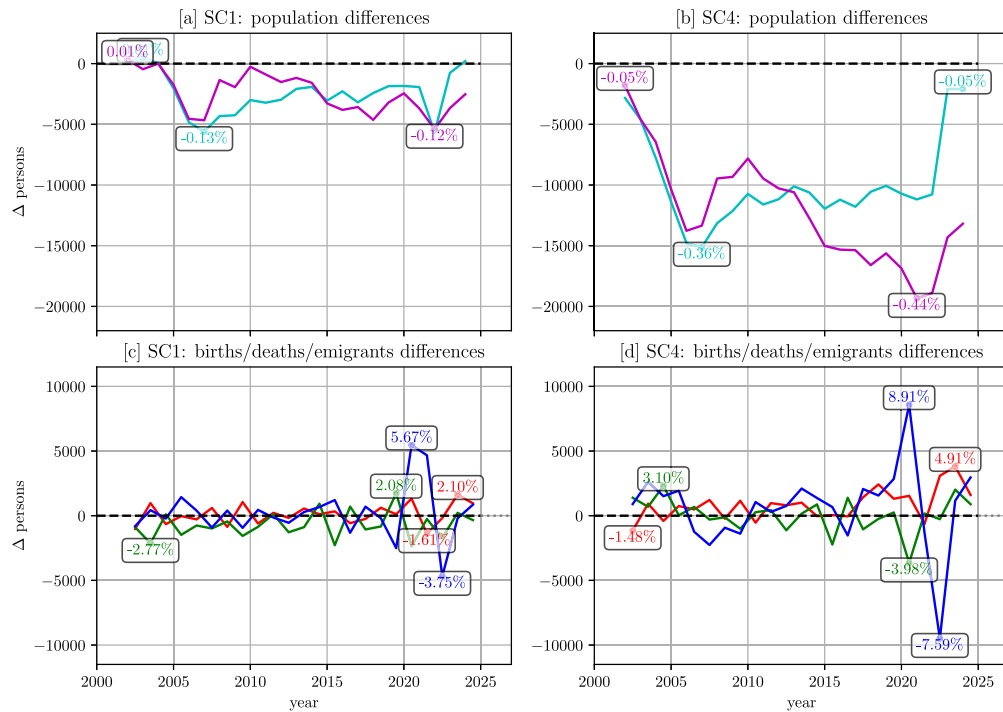


Fig. 8. Analogous to panels [c] and [d] of Fig. 5, focusing on the period between 2000 and 2025. The panels compare scenarios SC1, which uses formula (2) for probability computation, and SC4, which uses a naive computation.

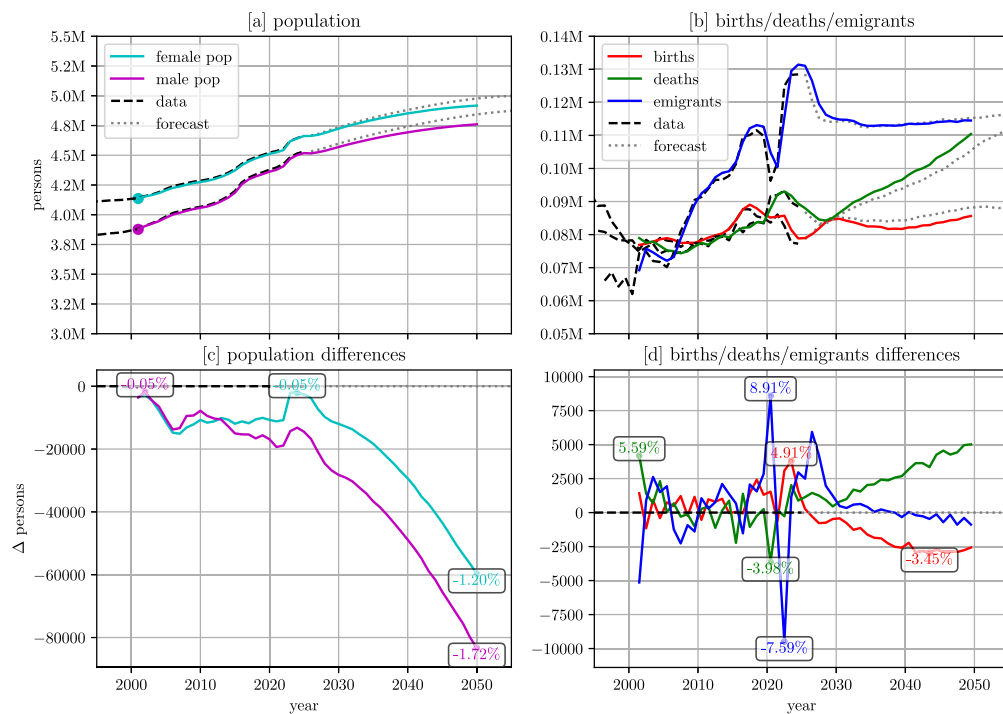


Fig. 9. Analogous to Fig. 5 for scenario SC4, which uses naive computation of parameters instead of formula (2).

answered reasonably well using at least daily macro steps – we recognise it as a deliberate design trade-off: interaction latency is bounded by the macro-step length and must be chosen case-by-case. Applications with highly timing-sensitive interactions or

complex causal chains between agents may instead require alternative synchronisation strategies, such as PDES-style optimistic synchronisation [46].

Birthday-based update scheme. Moreover, the agents' demographic update scheme, which is driven by *Birthday* events, presents a very intuitive embedding of demographic processes into any agent-based population model. The corresponding events schedule themselves dynamically; and are thus encapsulated and always active in the background. This allows modelling to focus on the processes relevant to the application, e.g. the modelling of contacts and disease dynamics in epidemiological applications. Moreover, it guarantees that the accuracy of the demographic computations, and to some extent also the computation time for the demographic computations, are independent of the chosen macro step size (see Table 2).

As mentioned earlier, this strategy also slightly improves the performance of the model, because every agent, at all times, precisely knows its current age in years. This may sound trivial, but a correct computation of an agent's age given the current date and the agent's birthdate may cause surprisingly high overheads (e.g. if the agent was born on a leap day).

On the downside, the update scheme still does not provide a solution to the presented chronometric dissonance problem (see Fig. 1). For perfect computation of demographic probabilities analogous to Definition 2.1, one would require a dataset with precise dates of demographic events and individual characteristics. Any computation from yearly census data must rely on approximations.

Algorithms for parameter calculation. The benchmarks shown in Figs. 7 and 8 show how important a sophisticated parameter calculation is for the model fit. Errors in the age-dependent parameters amplify themselves after multiple years of independent simulation. Therefore, well-thought-out parametrisation heuristics, such as the Farr formula or the IPF algorithm, are cornerstones of obtaining a model that fits the data reliably.

Nevertheless, research on improvement of parametrisation heuristics is still far from finished. For example, the differences visible in Fig. 5, panel [b], especially for the emigrants between 2020 and 2022, are not due to incomplete information, but exclusively due to imperfect parametrisation, and could be reduced with more accurate parametrisation formulas. Also, the disaggregation assumptions for the low-resolution projection data could be improved to better emulate the official population forecasts. The disaggregation and fitting heuristics also implicitly assume a degree of temporal stability of the underlying demographic structure; abrupt shocks such as sudden migration-regime changes or large policy reforms may reduce accuracy unless additional information is incorporated.

Due to its unique modelling purpose, it is difficult to compare the accuracy of GEPOC ABM with other models from the literature. For example, the developers of the Austrian STATSIM model report a root mean square error (RMSE) of 17 774 for the total population size when the model is run between 2013 and 2021 [21]. For the same period, GEPOC ABM has an RMSE of only approximately 7 500. Nevertheless, the comparison, however positive it may seem, is not fair due to the different modelling purposes of the two models: unlike STATSIM, GEPOC ABM was **not** developed for the purpose of population-forecasting, and therefore uses knowledge about actual population change between 2013 and 2021 to calculate its parameters. STATSIM instead uses extrapolation methods, such as the Lee–Carter method [47]. Since most demographic (microsimulation) models, such as STATSIM, have been developed for forecasting purposes, i.e. models of class [a] in the Related Work section, it is generally difficult to make a fair comparison. We can and must place higher demands on the accuracy of GEPOC ABM. Moreover, it is difficult to compare GEPOC ABM with the application-specific models from class [b] as well. Corresponding approaches either completely ignore population dynamics, such as GEMS [24], or model it in a much more simplified way without considering precise birthdates, e.g. MOSART [28]. Moreover, to the authors' knowledge, GEPOC ABM is the only comparable tool which supports (limited) continuous-time agent-based modelling.

5.1. Model limitations

The flexibility to apply the model independently of the application area is not without disadvantages. Each application must first be carefully checked for feasibility with regard to the general limitations of the underlying population model, which we want to summarise:

On the one hand, the model inherits the limitations of the parametrisation data, which refer not only to quality, but also to resolution. For example, due to the low resolution of the corresponding datasets, performing simulations far into the future using the open-data parametrisation is only reasonable if slight underestimation of the population compared to the official forecast is acceptable. On the other hand, the modelling assumptions also lead to limitations. In particular, the interaction strategy between the agents makes it important to use an adequate macro time-step length, which is reasonable for the specific application. If chosen too large, the time-lag for interaction and creation of new agents will be too long, rendering the results inaccurate. If chosen too small, computation time will suffer from various unnecessary agent calls. In general, the comparatively long computation time can be considered a particularly important limitation for GEPOC ABM. Depending on the complexity of the application, simulation times are more likely to be in the range of minutes or hours rather than seconds or microseconds. This drastically limits its usability within systematic optimisation or calibration loops. Finally, the mentioned static applications of the tool have different requirements for validity and therefore come with different limitations. While, for example, sampling of valid location coordinates is hardly important for the dynamic simulation, it is crucial for spatial analysis of a static population snapshot. Currently, we apply a strategy which combines uniform sampling from open census data within geometries (municipality regions) with open raster data (settlement map). While the concept provides a visibly accurate picture of the population distribution within Austria, one finds weaknesses when zooming in.

Table A.3

Relative minimum and maximum deviations between the simulated census and the reference data for the total population on January 1st between 2000 and 2050, aggregated along single dimensions (overall, by sex, by age class, and by federal state). The depth of the colour indicates by how much the values are under- or overestimated.

region	sex	age	e_{min} (90% UI)	e_{max} (90% UI)
–	–	–	–0.92% (–1.0, –0.9)	0.01% (–0.0, 0.0)
–	male	–	–1.06% (–1.1, –1.0)	0.01% (–0.0, 0.0)
–	female	–	–0.79% (–0.9, –0.7)	0.05% (0.0, 0.1)
–	–	[0, 19]	–1.85% (–2.0, –1.8)	0.93% (0.9, 1.0)
–	–	[20, 39]	–2.09% (–2.1, –2.0)	0.34% (0.3, 0.4)
–	–	[40, 59]	–2.63% (–2.7, –2.6)	–0.13% (–0.1, –0.1)
–	–	[60, 79]	–0.47% (–0.5, –0.4)	1.77% (1.7, 1.8)
–	–	80+	–1.67% (–1.8, –1.6)	0.67% (0.6, 0.8)
AT-1	–	–	–0.74% (–0.9, –0.6)	0.34% (0.1, 0.5)
AT-2	–	–	–1.67% (–1.8, –1.5)	–0.01% (–0.0, 0.0)
AT-3	–	–	–0.61% (–0.7, –0.5)	0.20% (0.1, 0.2)
AT-4	–	–	–1.49% (–1.6, –1.4)	0.25% (0.2, 0.3)
AT-5	–	–	–0.01% (–0.2, 0.1)	0.44% (0.3, 0.6)
AT-6	–	–	–1.72% (–1.8, –1.6)	–0.25% (–0.4, –0.1)
AT-7	–	–	–1.73% (–1.9, –1.6)	0.37% (0.3, 0.4)
AT-8	–	–	–2.08% (–2.3, –1.9)	0.19% (0.1, 0.2)
AT-9	–	–	–0.47% (–0.5, –0.4)	0.33% (0.3, 0.4)

5.2. Conclusion and outlook

In recent years, GEPOC ABM has proven itself not only in a wide variety of applications, but also in different application modes. Classic agent-based simulation, where complexity originates from interaction, e.g. as in [12], where GEPOC ABM was used as an agent-based epidemic model, is only one of these modes. Sometimes, the model is applied in a static version. In this case, the agent population is sampled and immediately exported for further processing. This leaves space for cross-sectional analysis, where the synthetic population is compared with other data, such as in [17], where it is compared with flooding data. Alternatively, the snapshot can also be used as input for other models, such as in [48], where a static GEPOC ABM export was used to generate a spatial network for a dynamic, differential-equation-based wastewater model. Finally, sometimes the model is applied dynamically but without interaction between the agents — i.e. the model is used as a microsimulation model. Examples are [35,49], where agents were enhanced by certain health-specific properties and dynamically updated without direct interaction. With these examples, we want to emphasise that applications of GEPOC ABM beyond the traditional use of agent-based simulation are not only possible, but can also be highly relevant for decision support.

While the innovations presented in this work played a major role in improving the model in recent years, there is still room for improvement, in particular with respect to the discussed limitations. On the one hand, we aim to improve assumptions for the computation of parameters in order to further reduce the deviations from the census data, using more accurate parametrisation formulas. Furthermore, an alternative sampling method based on building-level data instead of a settlement raster map is currently under development to improve the validity of static applications. Moreover, we aim to further reduce computation time. Current plans include the described concepts for parallelisation at agent level and testing of other algorithms and programming languages. First benchmarks with the programming language Julia as well as the development of C/C++ libraries with Python and Java bindings are currently in the planning phase.

In conclusion, we have presented various methods that have been primarily responsible for making GEPOC ABM more flexible, valid and efficient in recent years, and thus increasingly used in decision support. These methods are general and can be transferred to any agent-based model with appropriate adjustments.

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Appendix

Tables A.3 and A.4 show the differences between simulation results and source data on different levels of aggregation.

Data availability

The used data is openly available from a source cited in the text.

Table A.4

Relative minimum and maximum deviations between the simulated census and the reference data for the total population on January 1st between 2000 and 2050, broken down by federal state and age class. The depth of the colour indicates by how much the values are under- or overestimated.

region	sex	age	e_{min} (90% UI)	e_{max} (90% UI)
AT-1	–	[0, 19]	–2.59% (–2.9, –2.1)	1.19% (0.9, 1.6)
AT-1	–	[20, 39]	–3.87% (–4.3, –3.4)	0.97% (0.6, 1.4)
AT-1	–	[40, 59]	–4.42% (–4.7, –4.0)	0.89% (0.5, 1.3)
AT-1	–	[60, 79]	–0.45% (–0.6, –0.2)	4.58% (4.2, 5.0)
AT-1	–	80+	–1.80% (–2.6, –1.4)	1.91% (1.4, 2.5)
AT-2	–	[0, 19]	–2.38% (–2.7, –1.9)	0.78% (0.6, 1.0)
AT-2	–	[20, 39]	–0.77% (–1.1, –0.4)	0.42% (0.1, 0.8)
AT-2	–	[40, 59]	–2.69% (–2.9, –2.4)	–0.01% (–0.1, 0.1)
AT-2	–	[60, 79]	–1.31% (–1.6, –1.0)	–0.20% (–0.3, 0.1)
AT-2	–	80+	–3.06% (–3.5, –2.5)	–0.01% (–0.5, 0.4)
AT-3	–	[0, 19]	–3.24% (–3.5, –3.1)	0.83% (0.7, 1.0)
AT-3	–	[20, 39]	–3.44% (–3.6, –3.3)	0.49% (0.3, 0.7)
AT-3	–	[40, 59]	–2.57% (–2.7, –2.4)	0.27% (0.1, 0.4)
AT-3	–	[60, 79]	–0.42% (–0.5, –0.4)	4.36% (4.2, 4.5)
AT-3	–	80+	–1.84% (–2.2, –1.5)	1.69% (1.6, 2.0)
AT-4	–	[0, 19]	–3.63% (–4.0, –3.3)	0.83% (0.8, 0.9)
AT-4	–	[20, 39]	–3.20% (–3.4, –3.1)	0.63% (0.6, 0.7)
AT-4	–	[40, 59]	–2.85% (–3.0, –2.7)	0.63% (0.5, 0.8)
AT-4	–	[60, 79]	–0.60% (–0.7, –0.5)	2.40% (2.2, 2.6)
AT-4	–	80+	–2.45% (–2.8, –2.1)	0.06% (–0.2, 0.4)
AT-5	–	[0, 19]	0.14% (–0.2, 0.5)	2.66% (2.0, 3.3)
AT-5	–	[20, 39]	–3.16% (–3.5, –2.8)	0.43% (0.2, 0.7)
AT-5	–	[40, 59]	–1.11% (–1.5, –0.9)	0.66% (0.5, 0.8)
AT-5	–	[60, 79]	–0.63% (–0.7, –0.5)	2.24% (2.1, 2.4)
AT-5	–	80+	–2.23% (–2.7, –1.6)	1.64% (1.2, 2.2)
AT-6	–	[0, 19]	–2.89% (–3.1, –2.6)	0.71% (0.5, 1.0)
AT-6	–	[20, 39]	–3.61% (–3.8, –3.5)	0.35% (0.2, 0.6)
AT-6	–	[40, 59]	–3.61% (–3.8, –3.4)	–0.39% (–0.5, –0.3)
AT-6	–	[60, 79]	–0.60% (–0.8, –0.5)	1.47% (1.2, 1.6)
AT-6	–	80+	–2.70% (–3.0, –2.4)	0.40% (0.1, 0.8)
AT-7	–	[0, 19]	–4.86% (–5.2, –4.5)	1.14% (1.0, 1.2)
AT-7	–	[20, 39]	–2.96% (–3.3, –2.7)	0.32% (0.2, 0.5)
AT-7	–	[40, 59]	–2.62% (–2.9, –2.3)	0.30% (0.2, 0.3)
AT-7	–	[60, 79]	–0.33% (–0.5, –0.1)	1.29% (1.1, 1.5)
AT-7	–	80+	–1.40% (–1.9, –0.6)	1.28% (1.1, 1.4)
AT-8	–	[0, 19]	–3.88% (–4.4, –3.5)	0.60% (0.5, 0.8)
AT-8	–	[20, 39]	–1.78% (–2.2, –1.5)	0.38% (0.2, 0.7)
AT-8	–	[40, 59]	–4.16% (–4.5, –3.8)	0.16% (0.1, 0.4)
AT-8	–	[60, 79]	–0.37% (–0.6, –0.1)	0.29% (0.1, 0.6)
AT-8	–	80+	–2.58% (–3.0, –2.3)	0.81% (0.2, 1.6)
AT-9	–	[0, 19]	–0.73% (–1.0, –0.5)	3.21% (2.9, 3.4)
AT-9	–	[20, 39]	–0.36% (–0.4, –0.3)	1.37% (1.3, 1.5)
AT-9	–	[40, 59]	–1.91% (–2.1, –1.8)	–0.45% (–0.6, –0.3)
AT-9	–	[60, 79]	–0.72% (–0.9, –0.6)	–0.07% (–0.2, 0.1)
AT-9	–	80+	–2.03% (–2.3, –1.6)	1.45% (1.2, 1.8)

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